raised fibres all through its surface. The examples of napped fabrics are cotton flannel, rayon flannel, woollen and worsted napped fabric like kersey and melton. Napped fabrics have softer handle, better insulation properties due to more air entrapment. These fabrics are mainly used as blankets and for winter clothing.



Figure 1.10 Napping Machine

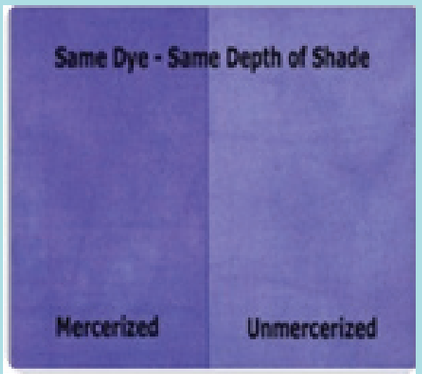
1.3.11 Mercerization

It is a finishing treatment of cotton and/or natural fibres composed by cellulose with a strong caustic alkaline solution (300 g/l) in order to improve its appearance by making it lustrous. The finish was named after its discoverer, John Mercer (1791-1866). He invented a process in which cotton can be given a lustrous finish resembling silk which was named "mercerization". The strong caustic soda on cellulosic material causes the fibres to swell and simultaneously there is a longitudinal shrinkage in the fibre. The morphological structure of the fibre gets modified giving it a shinier surface which is also resistant to wear and washing. Thus, we can say that mercerising results in the swelling of the cell wall of the cotton fibre which increases its surface area and reflectance and giving the fibre a softer feel.

DO YOU KNOW?

Why is mercerization been done on textile fabrics?

It is done to impart a greater affinity for dyes and chemical finishes, to increase tensile strength and lustre



Types of Mercerization

There are two types of mercerization:

Tension Mercerization

- The purpose of tension mercerization is to increase lustre of Cotton fibres
- The fibre untwists and swells, lumen becomes rounder in cross-section and it gains lustre
- Dye affinity and chemical reactivity increase
- Fabric becomes stronger and smoother

Slack Mercerization

- Slack mercerization is not as lustrous as tension mercerization
- Improves elongation and recovery properties
- Used for producing comfort stretch garments and fabric bandages, which need to conform to body shapes.

Mercerisation alters the chemical structure of the cotton fibre. The structure of the fibre inter-converts from alpha-cellulose to a thermodynamically more favourable beta-cellulose polymorph.

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